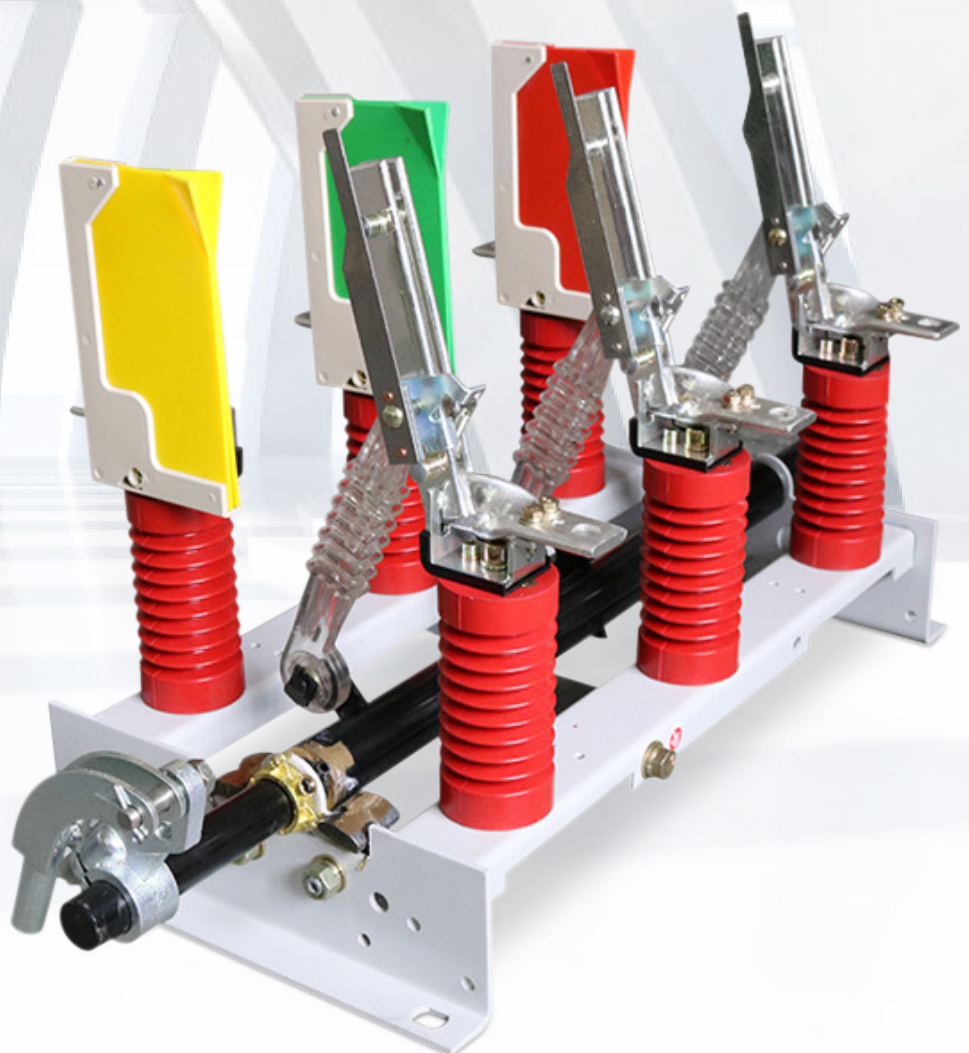




GreenPower

GPFN7 Series

Indoor High Voltage AC Load Switch Product Manual Book





About us »

GreenPower promotes environmental awareness, and aims to create unprecedented happiness and wealth for our investors, employees, clients and partners. By focusing in the mid-to-high voltage and low voltage eld, and professionally working on R&D, manufacturing, marketing and service of high-end green intelligent switchgears, equipments and products.GreenPower sets to become a well-respected global company in the power industry.

GreenPower, is jointly established by a number of state-owned excellent industrial electrical professional manufacturers, we committed to meet the procurement needs of clients. It is a professional procurement service provider with entity of industrial electrical.

GreenPower provides ONE-STOP solutions service for all customers in the International area. It is located in the time-honored electrical industrial city, a collection of worldclass brand of industrial electrical products,excellent brand of domestic electrical products as the basis.

Innovative business philosophy, strong professional supply team, expert technical guidance, advanced Information network management platform, fast logistics, which makes GreenPower as your procurement expert by your side.

GreenPower adheres to its own brand and multi-brand integration, marketing dierentiated development strategy. The products involving low voltage, medium voltage and high voltage transmission and distribution products and industrial automation products.

It covering all aspects of distribution, logistics, warehousing, professional and technical engineering services, systems integration and complete sets of manufacturing. Stable business foundation for cooperation with customers and suppliers, and has established a good reputation to maintain its leading position in the market competition.

Human quests for transcendence, with transcendence we make progress, when the pursuit of unlimited become a belief, our dream began to realize.

Hard-working, dedicated GreenPower people, will be adhering to its past glory, beginners mind, let go sailing, to face the fierce competition in the future, and always help customers make the best choice, and strive to become the best and most reliable procurement service provider in the eld of industrial electrical.



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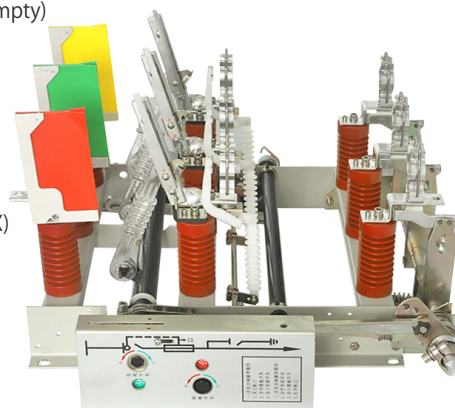
1. Main purpose and scope of application

GPFN7 AC high voltage load switch (hereinafter referred to as load switch) is a new type of gas-producing indoor high voltage load switch. It is suitable for three-phase AC power system with AC 50Hz and rated voltage of 12kV, used for breaking load current and closing short-circuit current.

2. Product model and meaning

GP FN 7 - 12 / C D □ □ R □ □ / □ □ □

- Rated current (400, 630A)
- Electric opening device (with: F, without: empty)
- Bumping fuse (with: A, without: empty)
- Fuse
- Load switch and earthing switch
- Interlock (with: L, without: empty)
- Earthing switch position
(Incoming terminal: S, Outgoing terminal: X)
- Earthing switch
- Switch side mounted
- Rated voltage (kV)
- Design Number
- Indoor load switch
- Company code, Green Power Electric



3. Normal working environment conditions

- a. Altitude: Less than 1000m;
- b. Ambient air temperature: Maximum +40°C , Minimum -25°C ;
- c. Relative humidity: The daily average is not more than 95%, and the monthly average is not more than 90%;
- d. The surrounding air should not be significantly polluted by corrosive or flammable gases and water vapor;
- e. No frequent violent vibrations.

4. Main specifications and technical parameters

Table 1 Main Specifications:

Name	Category	Full models	DS	DX	L	R	RA	F
			Earthing switch at the incoming end	Earthing switch at the outgoing end	Interlocking device	Fuse	Bumping fuse	Electric opening device
Load switch	Without trip unit	GPFN7-12	—	—	—	—	—	—
		GPFN7-12DSL	△	—	△	—	—	—
		GPFN7-12DXL	—	△	△	—	—	—
		GPFN7-12R	—	—	—	△	—	—
		GPFN7-12DSL R	△	—	△	△	—	—
		GPFN7-12DXLR	—	△	△	△	—	—
	With impact release	GPFN7-12RA	—	—	—	—	△	—
		GPFN7-12RAF	—	—	—	—	△	△
		GPFN7-12DXLRA	—	△	△	—	△	—
		GPFN7-12DXLRAF	—	△	△	—	△	△

Note: (—) without, (△) with.

Table 2 Rated parameters:

No.	Items	Unit	Data	
1	Rated voltage	kV	12	
2	Max. voltage	kV	12	
3	Rated current	A	400	630
4	Power frequency withstand voltage (1min)	kV	42/48	42/48
5	4S Thermal stable current (RMS)	kA	12.5	20
6	Dynamic stable current (peak value)	kA	31.5	50
7	Short-circuit making current	kA	31.5	50
8	Rated breaking current	A	400	630
9	Rated transfer current	A	1000	1000

Table 3 Fuse rated parameters:

Model	Rated voltage (kV)	Rated current (A)	Rated current of fuse (A)
SDLA*J	12	40	6.3, 10, 16, 20, 25, 31.5, 40
SFLA*J	12	100	50, 63, 71, 80, 100
SKLA*J	12	125	125

A: With impactor.

5. Structure overview

The load switch consists of base, insulator, moving and fixed contact knife, arc extinguishing knife, arc contact and arc extinguishing chamber, transmission mechanism and spring energy storage mechanism, fuse and CS6-1 or disc operating mechanism, etc., which can be connected to earthing switch. The switch is integrated and has an interlocking device.

6. Function

- Closing operation:
When manually closing, the closing spring is charged by the operating mechanism and the transmission mechanism. When the energy storage is terminated, the energy is released by the bounce to realize the high-speed closing of the contact knife.
- Opening operation and arc extinguishing principle:
When manually opening, the energy is released by the opening spring, so that the main contact knife is disconnected first, and then the arc contact knife breaks the circuit instantly.
At the moment when the arc contact knife is disconnected, an arc is generated, and the wall of the arc extinguishing chamber plays the role of cooling the arc, and generates gas under the action of the arc to extinguish the arc.
- The load switch equipped with a current-limiting fuse with a striker has an automatic tripping device. When the fuse is not blown in all phases, the load switch will automatically open, and can only be reclosed after re-buckling.
- The load switch equipped with the grounding switch is interlocked through the interlocking mechanism to avoid mis-opening and mis-closing.
- When the fuse is not matched with the integral load switch, the split load switch can be used.
- The load switch with automatic tripping device can be equipped with an electric opening device to realize electric opening.

7. Outline dimensions and installation

- The external dimensions are shown in Figures 1~8, and the installation method of the CS6-1 disc operating mechanism can be seen in Figures 9~12;
- Load switch with electric opening device, its electrical schematic diagram can be seen in Figure 13.

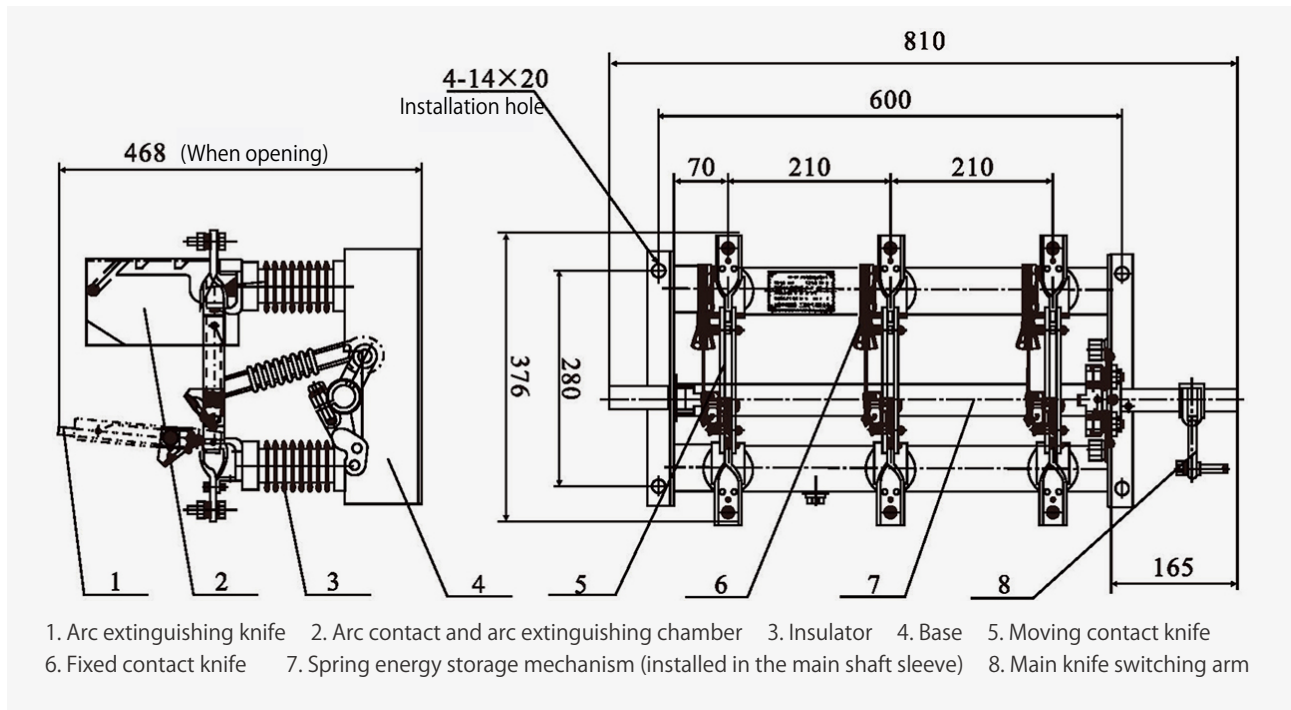


Figure 1 GPFN7-12 Load switch outline and installation dimensions

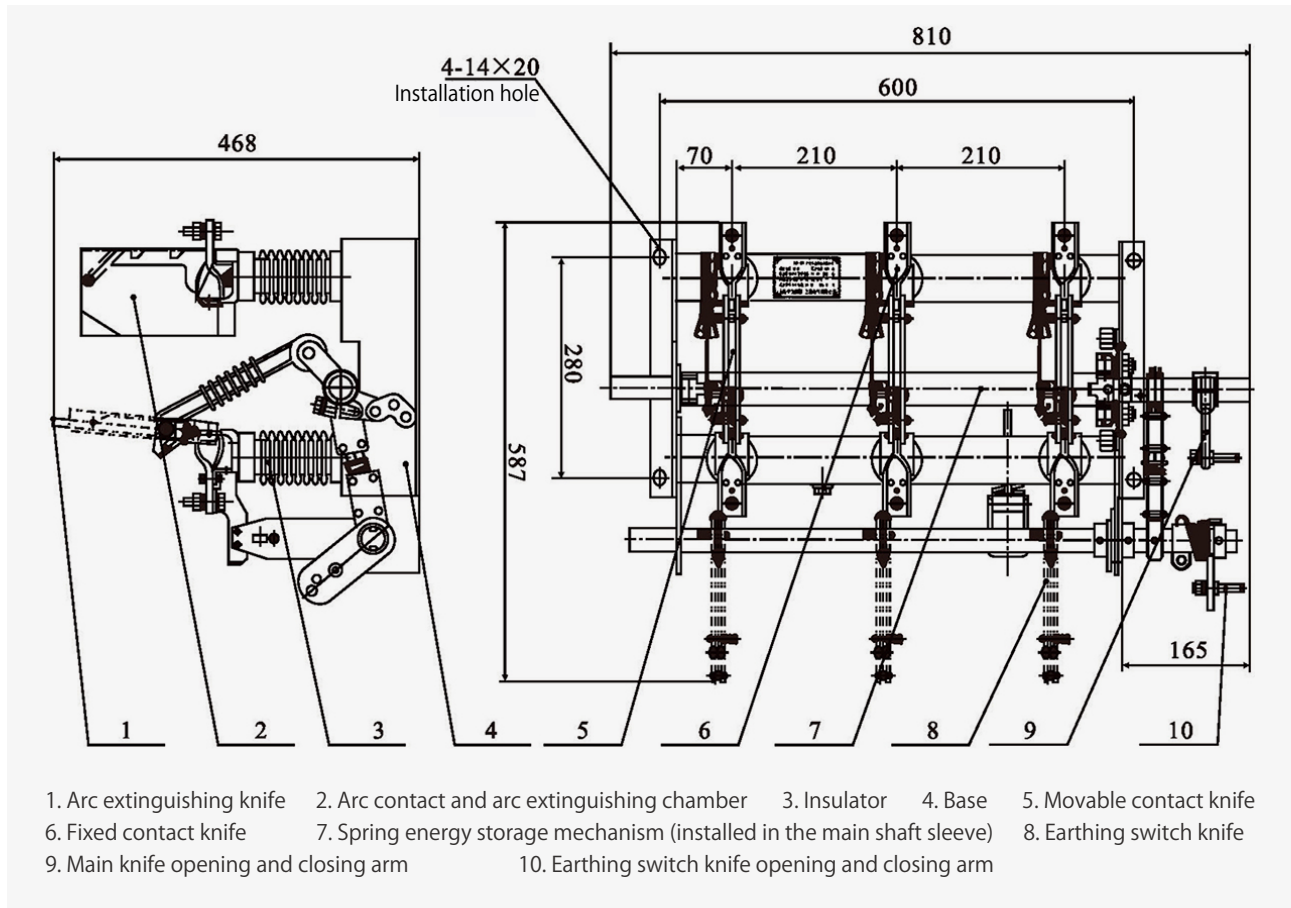


Figure 2 GPFN7-12DXL Load switch outline and installation dimensions

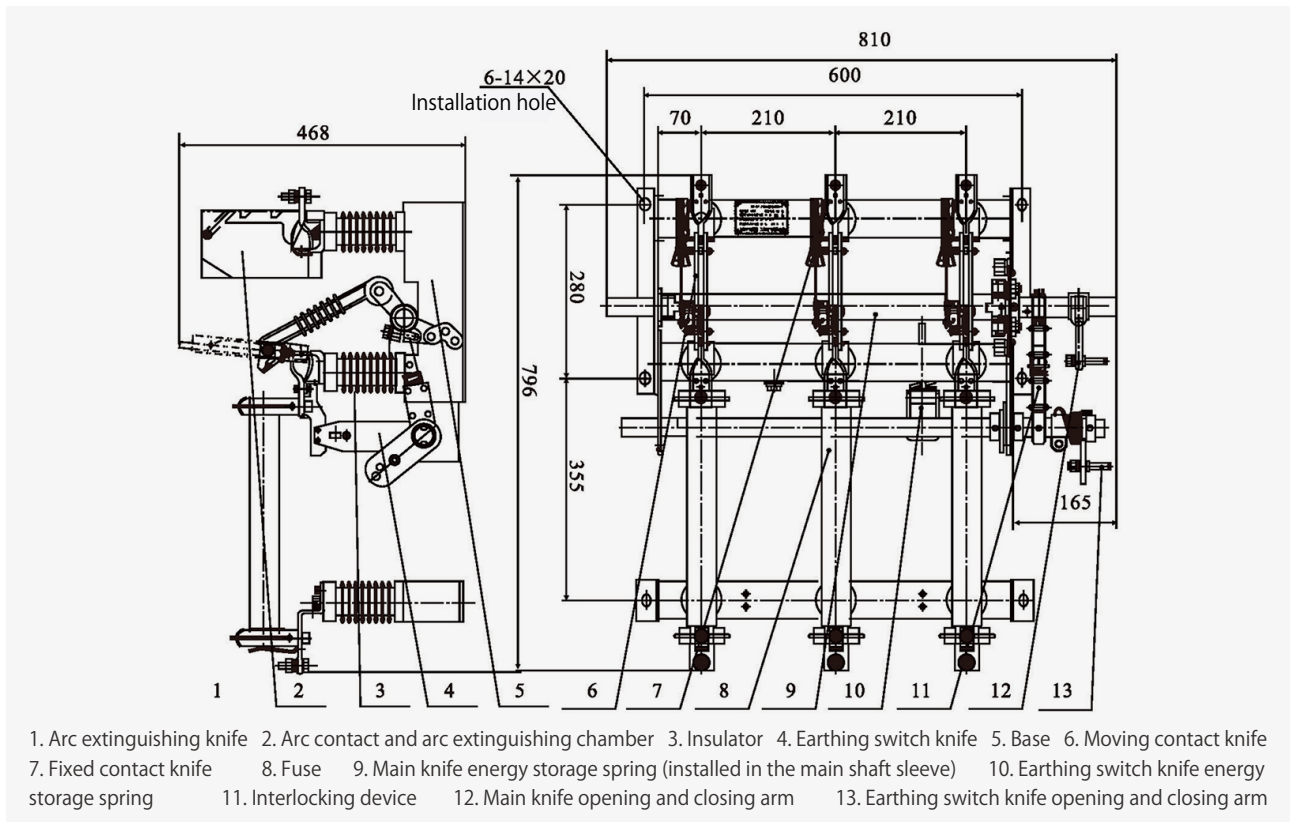


Figure 3 GPFN7-12DXLR Split load switch outline and installation dimensions

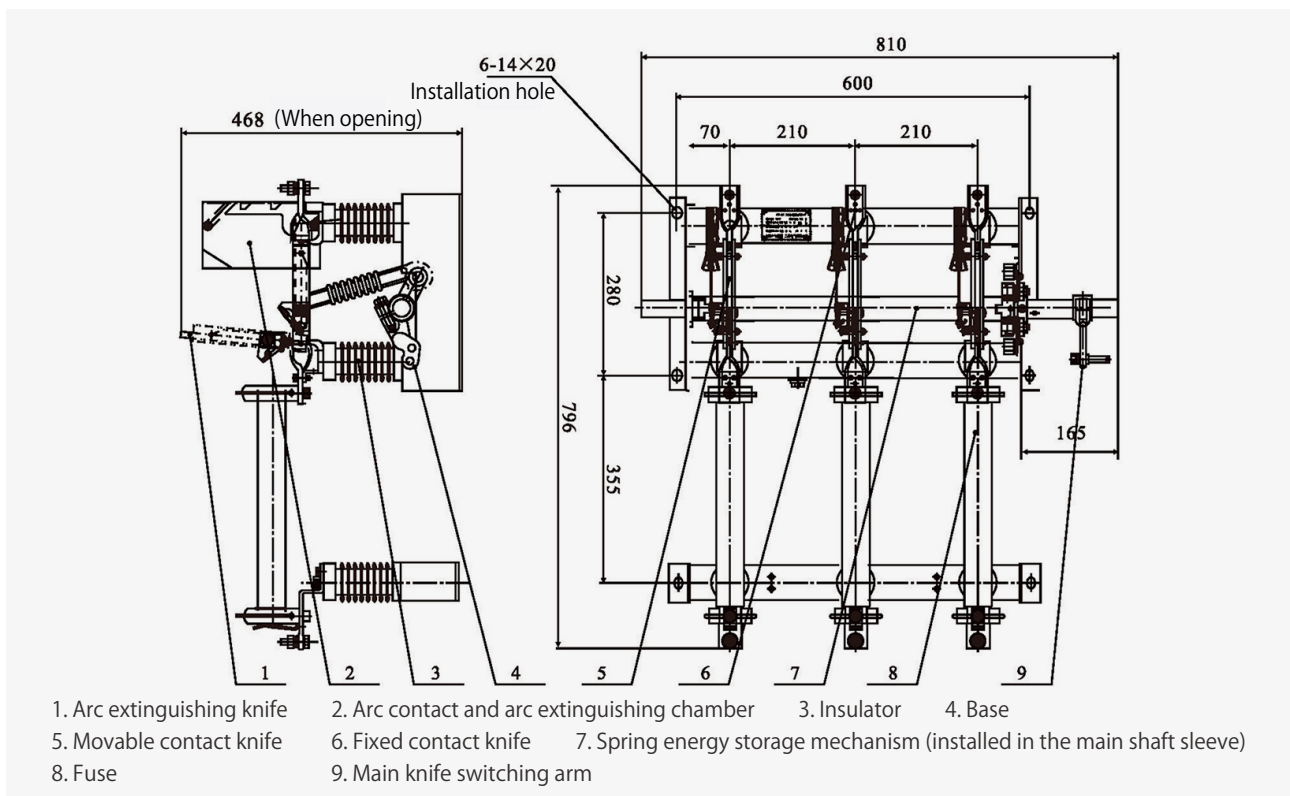


Figure 4 GPFN7-12R Split load switch outline and installation dimensions

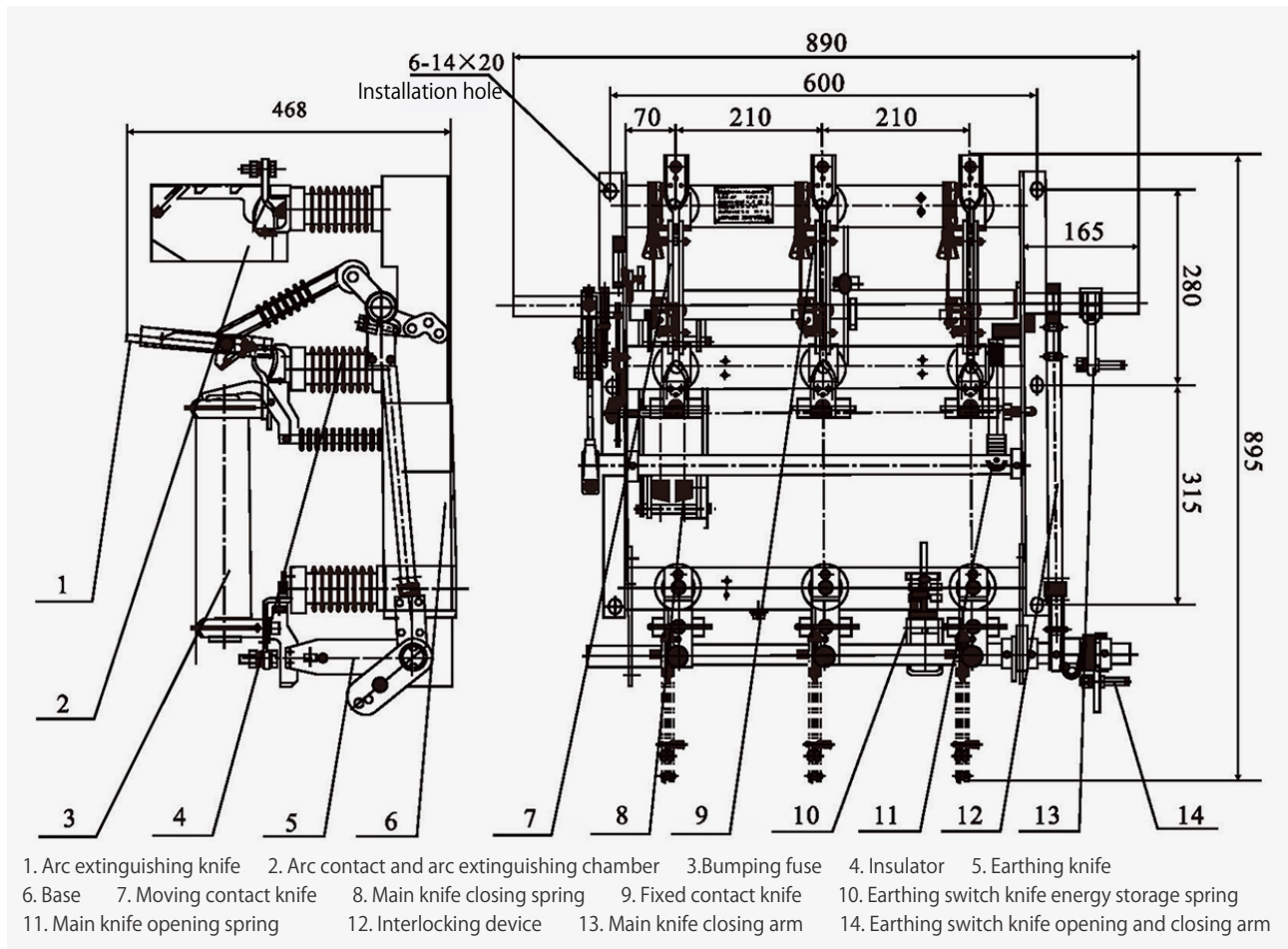


Figure 5 GPFN7-12DXLRA Integral load switch outline and installation dimensions

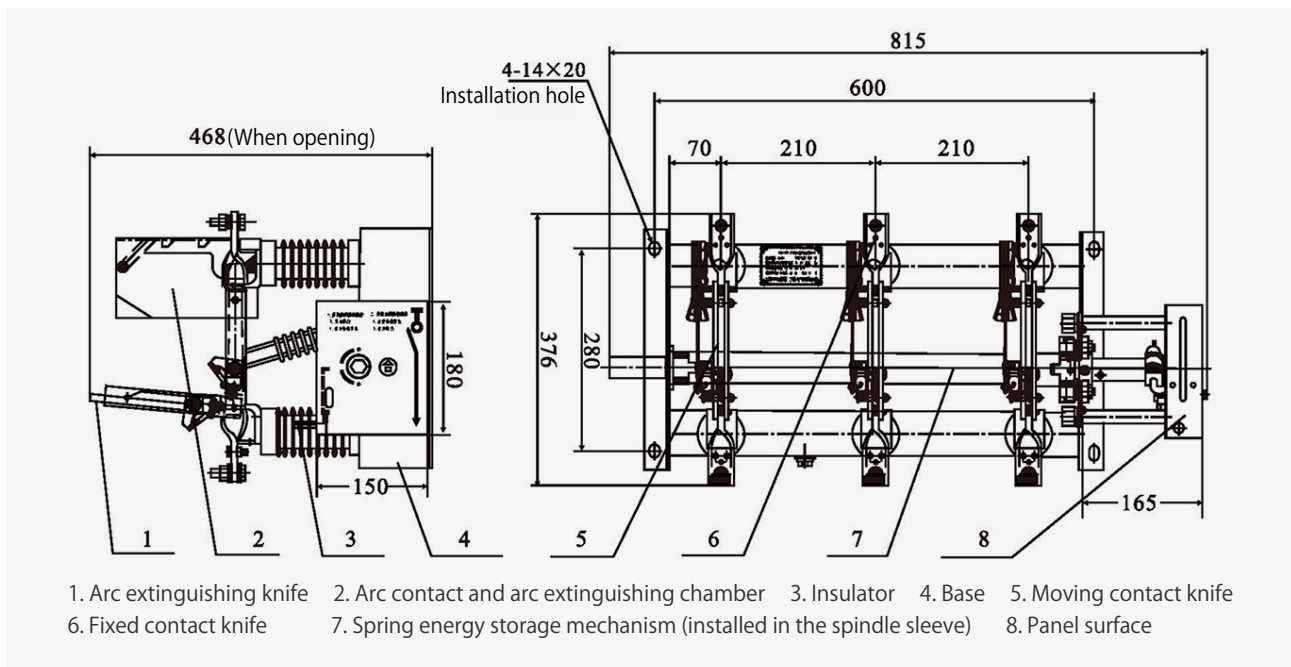


Figure 6 GPFN7-12C Load switch outline and installation dimensions

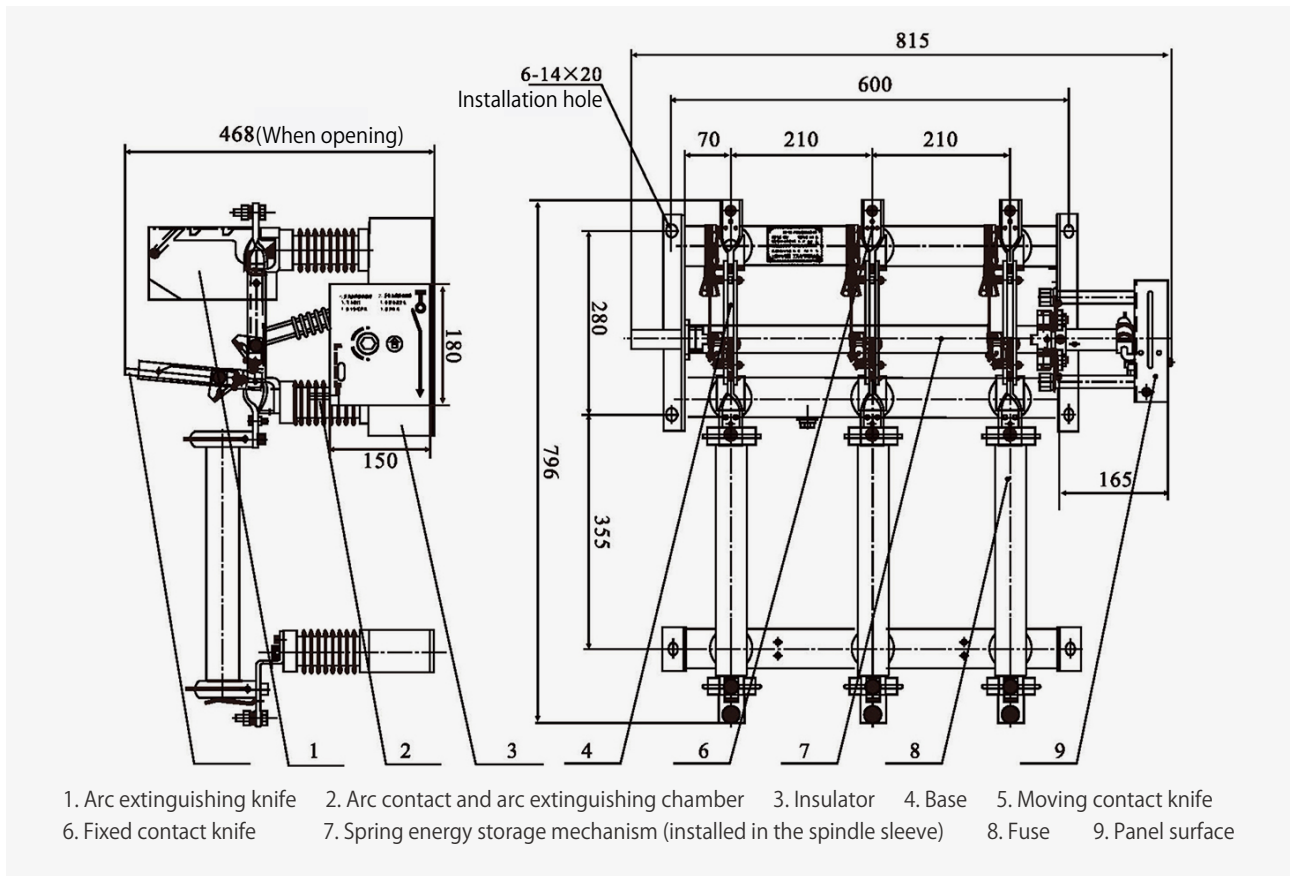


Figure 7 GPFN7-12CR Split load switch outline and installation dimensions

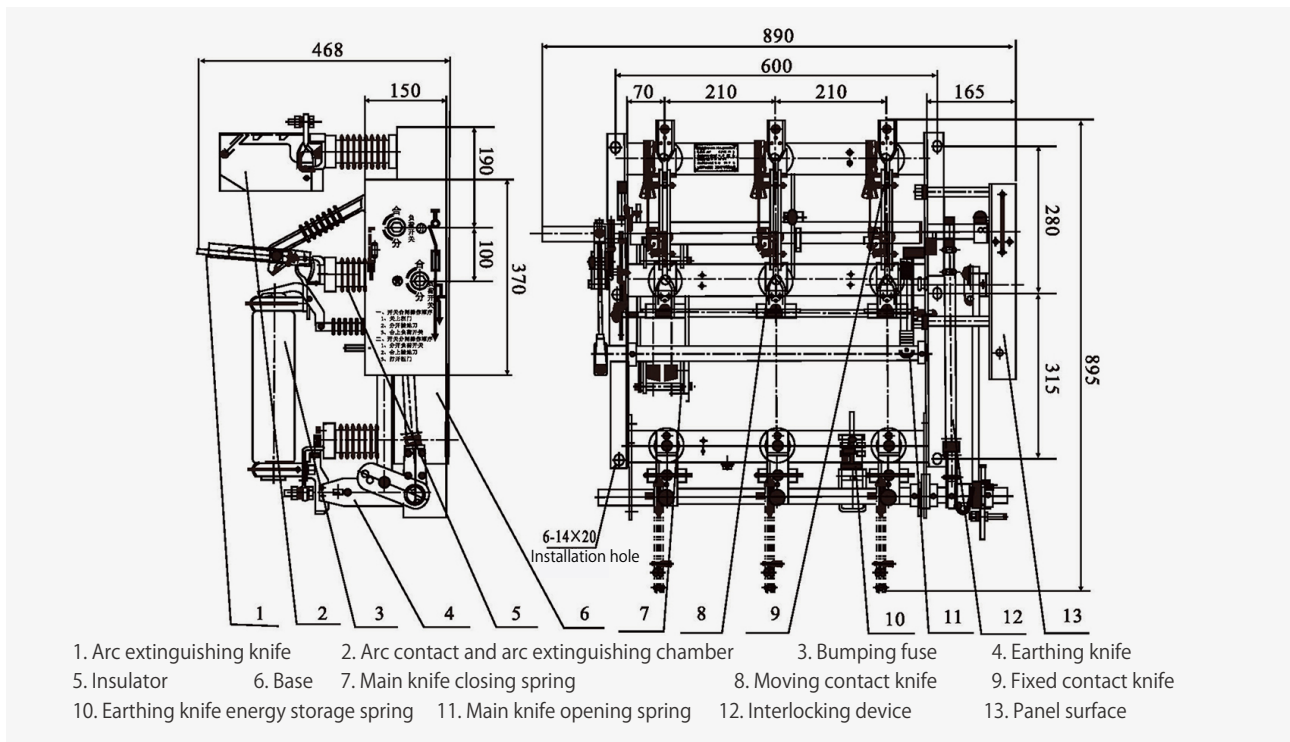


Figure 8 GPFN7-12CDXLRA Integral load switch outline and installation dimensions

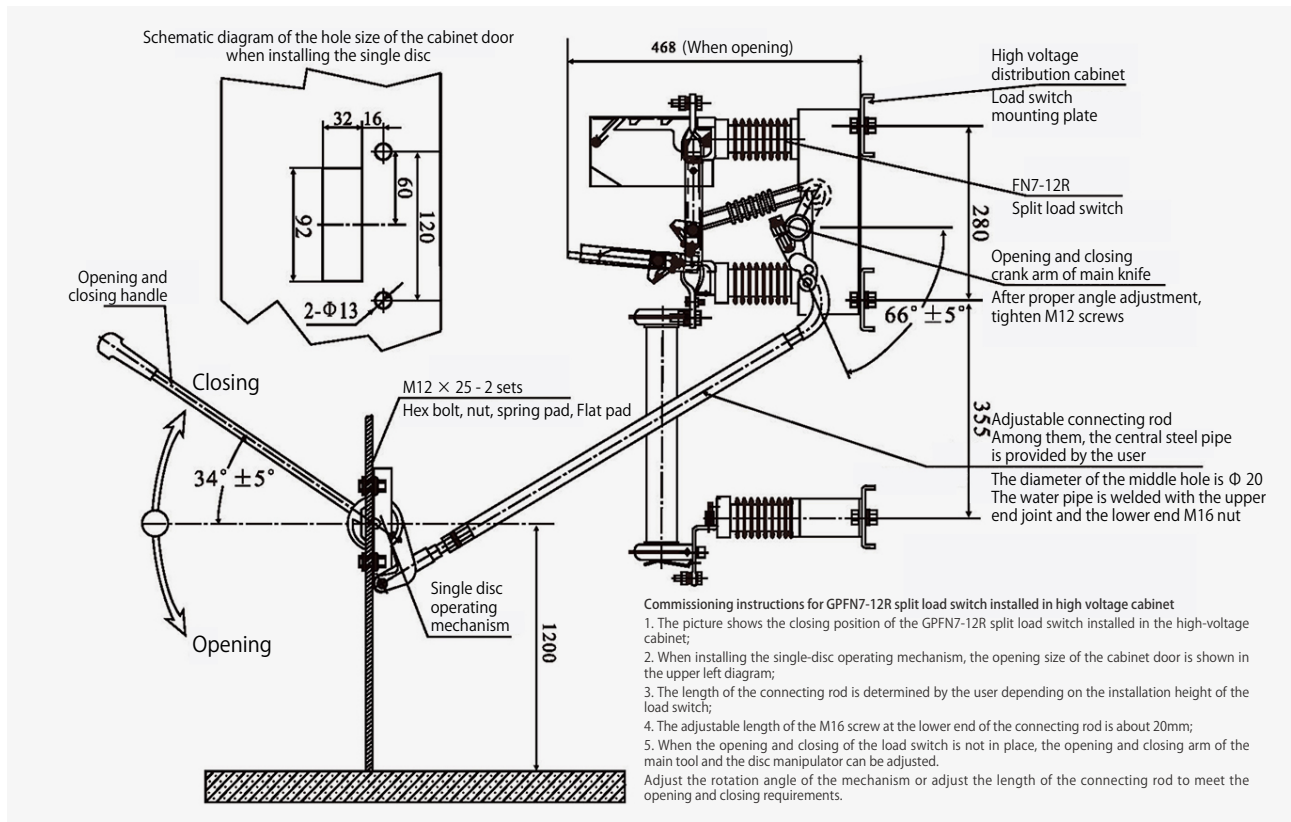


Figure 9 Schematic diagram of GPFN7-12 load switch and GPFN7-12R split type load switch installed in the cabinet

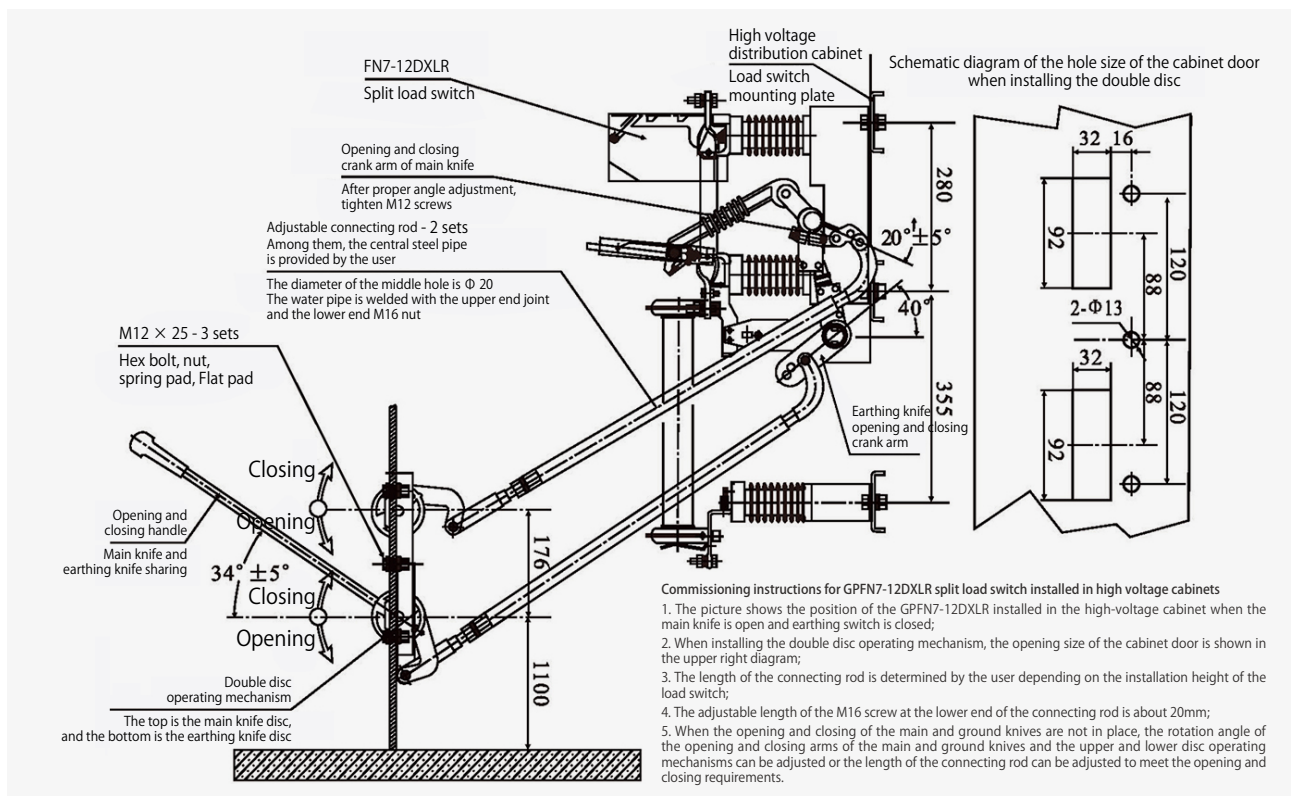


Figure 10 Schematic diagram of GPFN7-12DXLR split-type load switch installed in the cabinet

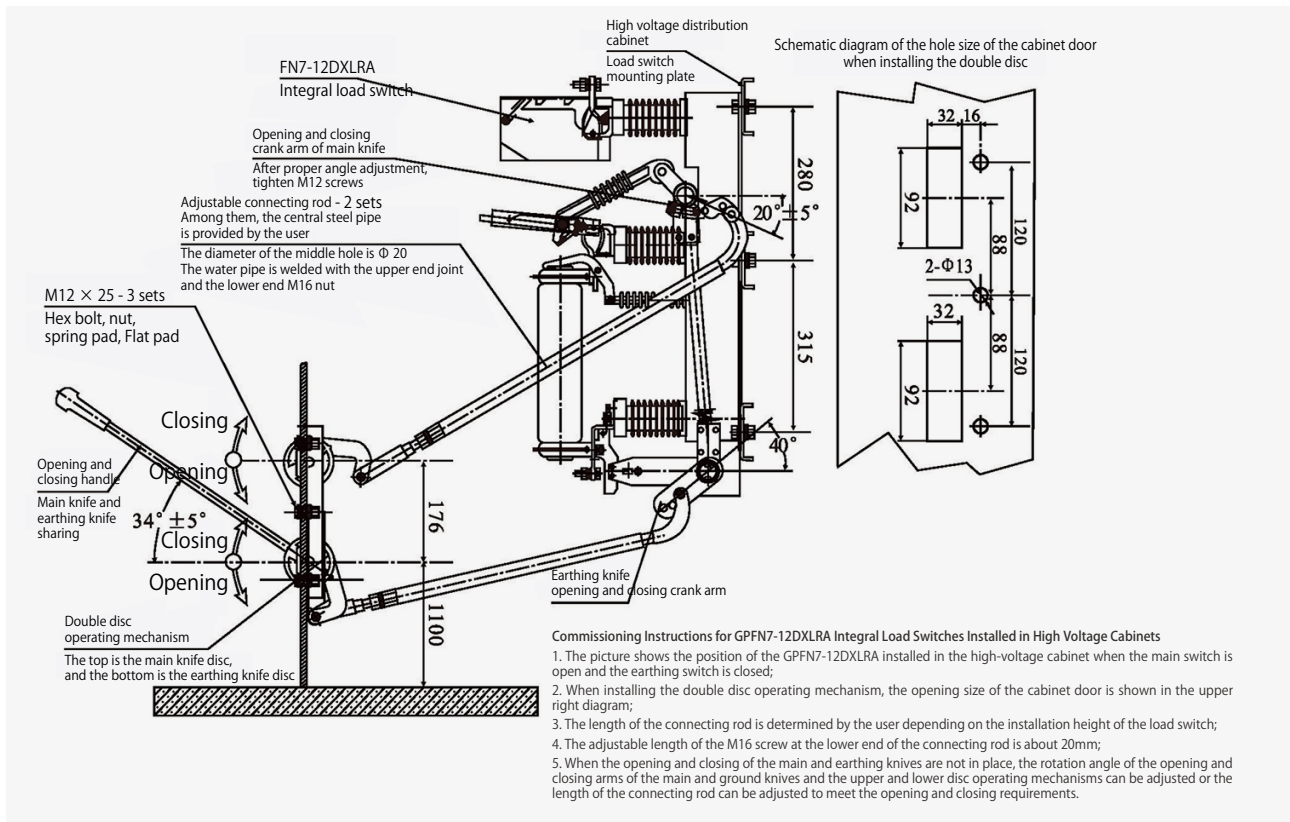


Figure 11 Schematic diagram of GPFN7-12DXLRA integral load switch installed in the cabinet

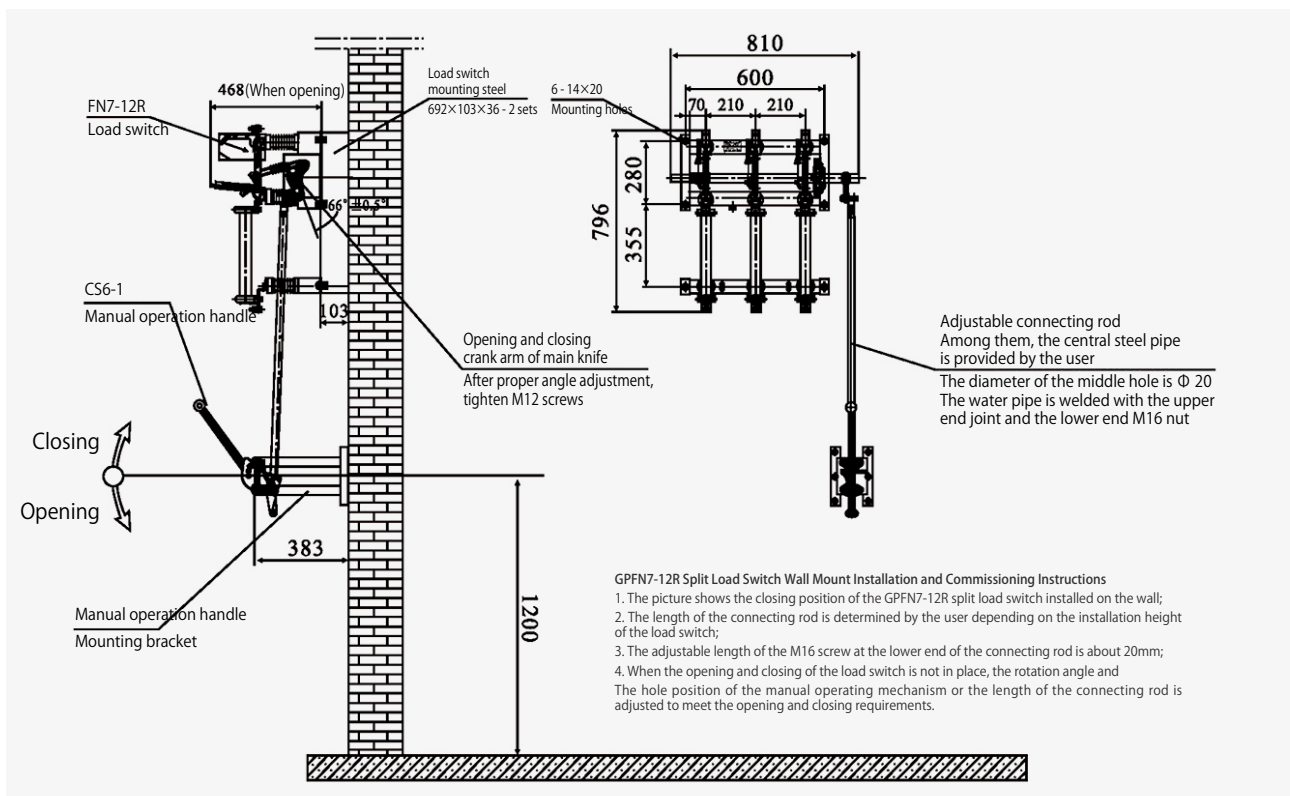
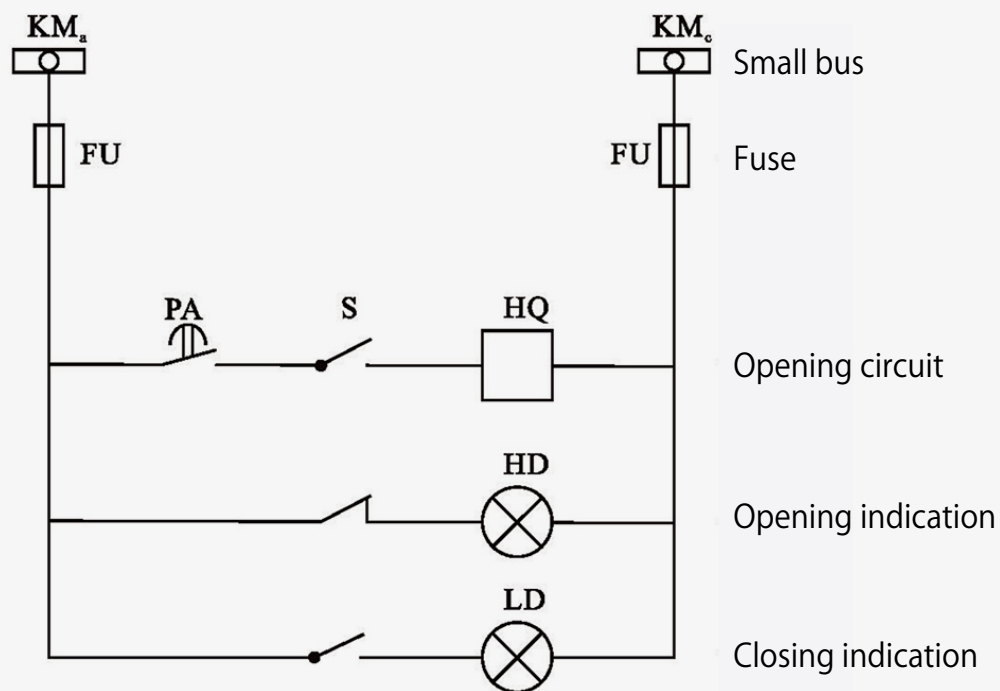


Figure 12 Wall-mounted installation of GPFN7-12 load switch and GPFN7-12R split load switch



No.	Code		Name	Model and specification	Qty.	Remarks
1	FU		Fuse	FS-10.5A	2	User-provided
2	PA		Open button	TE-XB2-EA	1	User-provided
3	S		Auxiliary switch	F9-4III/L AC 220V 5A	1	2K2B
4	HQ		Opening electromagnet	AC 220V	1	3kg force
5	HD	LD	Indicator	TE-XB2-EA	1	User-provided

Figure 13 The electrical schematic diagram of the GPFN7-12 load switch

8. Installation and commissioning

- a. The product has been debugged and passed the inspection before leaving the factory. Do not adjust and operate the switch without authorization before installation;
- b. The load switch should be installed on a vertical wall or metal frame, with 4~6 pcs M12 bolts, fixed at the installation hole position;
- c. Connect the switch busbars and copper bars. Before installation, they should be leveled and fastened with bolts to prevent the switch from being damaged or deformed by excessive force and affecting conduction and contact;
- d. The switch is equipped with CS6 type or disc manual operating mechanism. When the handle is up, it is the closing position, and when the handle is down, it is the opening position;
- e. After the load switch operating mechanism and the busbar are installed, the opening and closing operations should be performed no less than 5 times. The moving parts should be flexible, the conductive contact should be good, and the dynamic and fixed contact blades should not be dislocated due to the installation of the busbar.

9. Use and maintenance

- a. After unpacking, check whether the product and random technical documents are complete according to the packing list, and check whether the product is damaged during transportation and storage;
- b. Remove the dust from the product, and wipe the insulator and earthing switch parts clean.
- c. The conductive parts and mechanical friction parts are coated with neutral vaseline;
- d. After installation, check whether the wiring is correct and good, and whether the earthing is reliable.
- e. Regular inspection and maintenance should be carried out during operation, once a year; after breaking the rated load 20 times, a comprehensive overhaul should be carried out.
- f. The overhaul should be carried out without electricity, and the insulators and contacts should be checked.
- g. Operators must be familiar with all parts of the load switch, including performance, structure and installation.

10. Random document

- a. Product routine;
- b. Installation manual;
- c. Packing list.

11. Ordering notice

- a. Load switch model, name and quantity;
- b. If equipped with a fuse, please indicate the fuse model and its rated current.

Zhejiang Greenpower Electric Co., Ltd
Zhejiang GreenPower I&E Co.,Ltd
Professional Power Equipment provider

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